

5)

$$a) \int (\sin(3z + 4) + \frac{1}{z-2}) dz$$

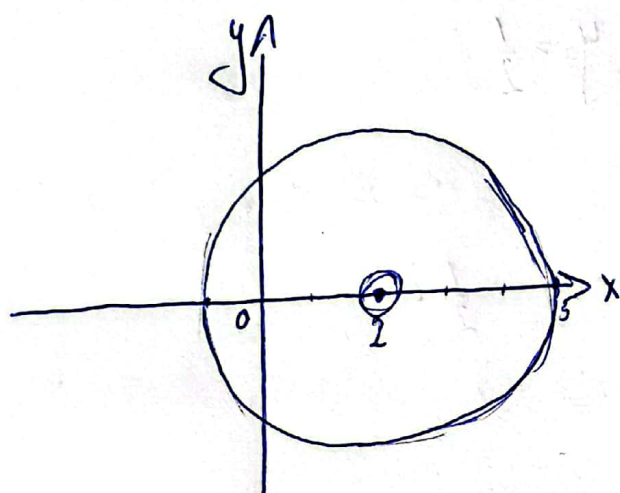
$$|z-2| = 3$$

$$C: |z - (2)| = 3$$

$$|z - z_0| = r$$

$$z_0 = 2$$

$$r = 3$$



$\sin(3z + 4) \rightarrow$ pot. trigonometrică de menteră $\Rightarrow$
 $\Rightarrow$ de morfă pe domeniu delimitat de curba C

$$z - 2 = 0 \Rightarrow z = 2 \Rightarrow A(2, 0) \in D \Rightarrow \text{pol de ord. } 1$$

$$f(z) = \frac{1}{2\pi i} \cdot \underbrace{\int_{(C_1)} \frac{f(z)}{z-2} dz}_{\mathcal{I}_1}$$

$$\mathcal{I} = \underbrace{\int_{(C_0)} \sin(3z + 4) dz}_{=0 \text{ din th. Concluz pt. domeniu simplu conex}} + \int_{(C_1)} \frac{1}{z-2} dz$$

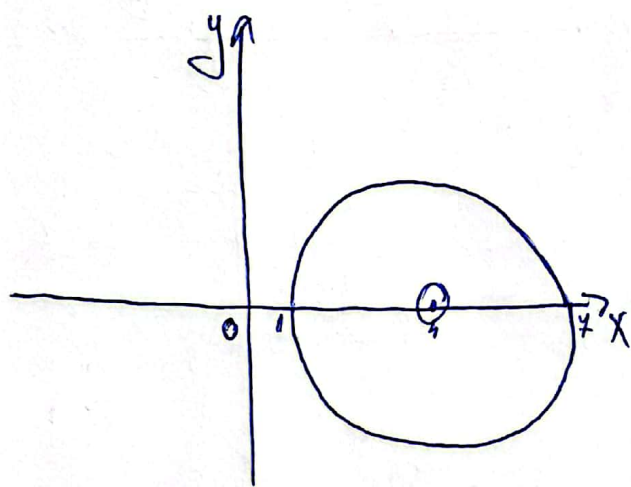
$= 0$ din th. Concluz pt. domeniu simplu conex

$$\frac{f(z)}{z-2} = \frac{1}{z-2} \Rightarrow f(z) = 1 \Rightarrow f(z) = 1(z)$$

$$(z) f(z) = 1 \Rightarrow 1 = \frac{1}{2\bar{h}i} \cdot \gamma \Rightarrow \gamma = 2\bar{h}i$$

$$b) \int_{|z-4|=3} \left(\cos(3z+4) + \frac{3z+4}{z-4} \right) dz$$

$$\begin{aligned} |z-4| &= 3 \\ |z-z_0| &= r \end{aligned} \quad \Rightarrow \quad \begin{aligned} z_0 &= 4 \\ r &= 3 \end{aligned}$$



$\cos(3z+4) \rightarrow$ pot. trigonometrică elementară $\Rightarrow$
 $\Rightarrow$ olemorfă pe $\mathbb{C} \Rightarrow$ olemorfă pe domeniul de-
 limitat de curba C

$$\gamma = \underbrace{\int_{(C_1)} \cos(3z+4) dz}_{\gamma_1 = 0 \text{ din tl. lni}} + \underbrace{\int_{(C_2)} \frac{3z+4}{z-4} dz}_{\gamma_2}$$

Conchy

$$z-4=0 \Rightarrow z=4 \quad A(4,0) \rightarrow \text{pol de ord. } 1$$

$$a=4$$

$$\Rightarrow f(4) = \frac{1}{2\pi i} \underbrace{\int_{(C_2)} \frac{f(z)}{z-4} dz}_{J_2}$$

$$\frac{f(z)}{z-4} = \frac{3z+4}{z-4} \Rightarrow f(z) = 3z+4$$

$$f(4) = 12+4=16$$

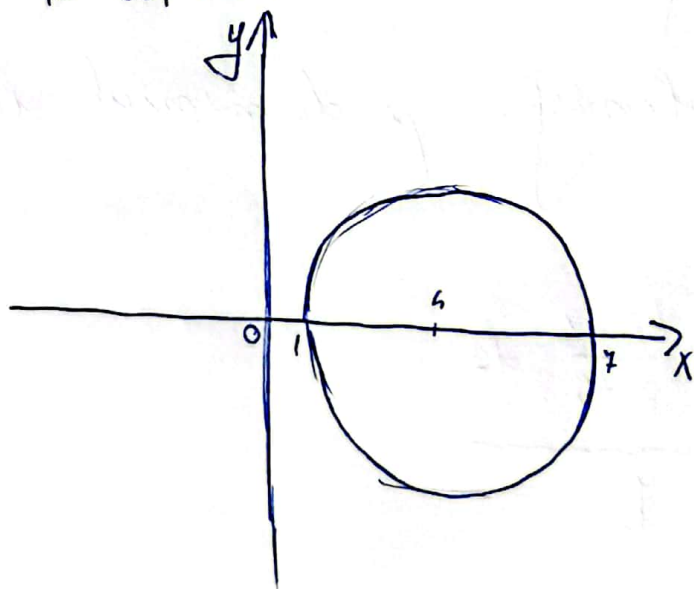
$$\Rightarrow 16 = \frac{1}{2\pi i} J_2 \Rightarrow J_2 = 32\pi i$$

$$\Rightarrow J = 32 \cdot \pi \cdot i$$

$$c) \int_{|z-4|=3} \frac{2z+5}{z^2-2z-3} dz$$

$$|z-4|=3$$

$$|z-z_0|=r \quad \Rightarrow z_0=4, \quad r=3$$



$$z^2 - 2z - 3 = 0$$

$$\Delta = 4 + 12 = 16$$

$$z_{1,2} = \frac{2 \pm 4}{2}$$

$$\Rightarrow \begin{cases} z_1 = 3 & B(3, 0) \in D \rightarrow \text{pol de ord. 1} \\ z_2 = -1 & A(-1, 0) \notin D \end{cases}$$

$$J = \int_C \frac{2z + 5}{z^2 - 2z - 3} dz = \int_C \frac{\frac{2z+5}{z+1}}{z-3} dz$$

$$z - 3 = 0 \Rightarrow z = 3$$

$$a = 3 \Rightarrow f(3) = \frac{1}{2\pi i} \int_{(C)} \frac{f(z)}{z-3} dz$$

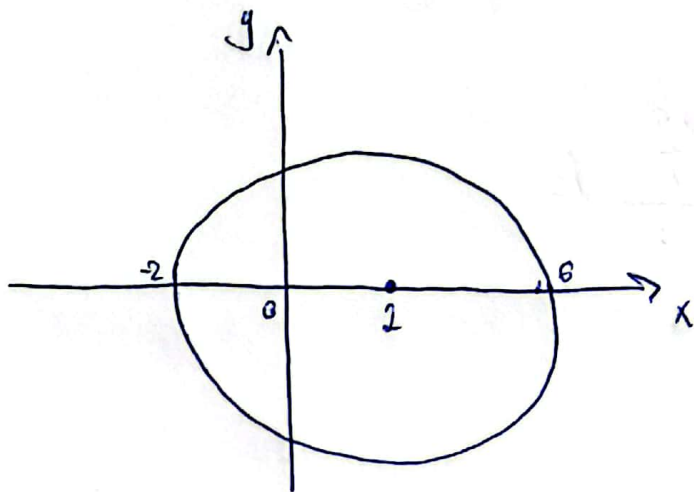
$$\frac{f(z)}{z-3} = \frac{2z+5}{\frac{z+1}{z-3}} \Rightarrow f(z) = \frac{2z+5}{z+1}$$

$$f(3) = \frac{6+5}{3+1} = \frac{11}{4}$$

$$\Rightarrow \frac{11}{4} = \frac{1}{2\pi i} \cdot J \Rightarrow J = \frac{11 \cdot 2\pi i}{4} = \frac{11\pi i}{2}$$

$$d) \int_{|z-2|=4} \frac{3z+4}{z^2(z^2+5z)} dz$$

$$\begin{aligned} |z-2|=4 \\ |z-z_0|=r \quad | \Rightarrow z_0=2, \quad r=4 \\ |z-z_0|=r \end{aligned}$$



$$z^2(z^2+5z) = 0 \Leftrightarrow z^3(z+5) = 0 \Rightarrow$$

$$\begin{aligned} \Rightarrow z_1 = z_2 = z_3 = 0 \Rightarrow A(0,0) \in D \rightarrow \text{pol de ord. III} \\ z+5=0 \Rightarrow z_4 = -5 \Rightarrow B(-5,0) \notin D \end{aligned}$$

$$\begin{aligned} z=0 \\ a=0 \quad \text{pol de ordim } m+1=3 \Rightarrow m=2 \end{aligned}$$

$$\Rightarrow \int_{(0)}^{(2)} = \frac{2!}{2\pi i} \int \frac{f(z)}{z^3} dz$$

$$\frac{f(z)}{z^3} = \frac{3z+4}{z^3(z+5)} \Rightarrow f(z) = \frac{3z+4}{z+5}$$

$$f'(z) = \frac{3 \cdot (z+5) - (3z+4) \cdot 1}{(z+5)^2} = \frac{3z + 15 - 3z - 4}{(z+5)^2} = \frac{11}{(z+5)^2}$$

$$f''(z) = \frac{0 - 11 \cdot 2(z+5) \cdot 1}{(z+5)^3} = -\frac{22}{(z+5)^3}$$

$$f''(0) = -\frac{22}{5^3} = -\frac{22}{125}$$

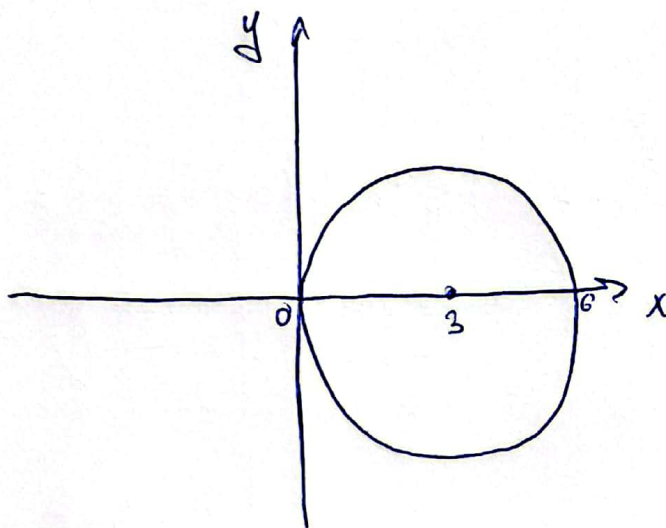
$$\Rightarrow \frac{-22}{125} = \frac{f}{2\pi i} \cdot \gamma \Rightarrow \gamma = \frac{-22\pi i}{125}$$

$$e) \int \frac{z+3}{z^2(z^2-5z)} dz$$

$|z-3|=3$

$$|z-3|=3$$

$$|z-z_0|=R \Rightarrow z_0=3, R=3$$



$$z^2(z^2-5z)=0 \Rightarrow \begin{cases} z_1=z_2=z_3=0 \Rightarrow A(0,0) \in D \rightarrow \text{pol de ord. 3} \\ z_4=5 \Rightarrow B(5,0) \in D \rightarrow \text{pol de ord. 1} \end{cases}$$

$$\gamma = \int_{C_1} \frac{z+3}{z^3} dz$$

$$a=0, m-1=3 \Rightarrow m=2$$

$$f''(0) = \frac{2!}{2\pi i} \int_{(C_1)} \underbrace{\frac{f(z)}{z^3}}_{\gamma_1} dz$$

$$\frac{f(z)}{z^3} = \frac{z+3}{z-5} \Rightarrow f(z) = \frac{z+3}{z-5}$$

$$f'(z) = \frac{1(z-5) - (z+3) \cdot 1}{(z-5)^2} = \frac{\cancel{z}-5-\cancel{z}-3}{(z-5)^2} = \frac{-8}{(z-5)^2}$$

$$f''(z) = \frac{0 - (-8) \cdot 2 \cdot (\cancel{z}-5) \cdot 1}{(z-5)^3} = \frac{16}{(z-5)^3}$$

$$f''(0) = \frac{16}{(0-5)^3} = -\frac{16}{125}$$

$$\gamma_1 = \pi i \cdot \left(-\frac{16}{125}\right) = -\frac{16\pi i}{125} \quad (1)$$

$$\gamma_2 = \int \frac{z+3}{z-5} dz$$

$$z = 5$$

$$f(5) = \frac{1}{2\pi i} \int_{(C_2)} \underbrace{\frac{f(z)}{z-5}}_{\gamma_2} dz$$

$$\Rightarrow f(z) = \frac{z+3}{z-5} \quad \gamma_2$$

$$f(5) = \frac{5+3}{125} = \frac{8}{125}$$

$$\Rightarrow \gamma_2 = 2\pi i \cdot \frac{8}{125} = \frac{16\pi i}{125} \quad (2)$$

$$\text{Dim } \textcircled{1} \quad \hat{s}_1 \quad \textcircled{2} \Rightarrow J = J_1 + J_2 = \frac{16\bar{u}\bar{i}}{125} - \frac{16\bar{a}\bar{i}}{125} = 0$$

6)